

Blinkit Sales Analysis

SQL & MySQL · Data Analysis Project

[Your Name] · [Date] · MySQL 8.x

Key Performance Indicators

\$1.20M Total Sales	\$141 Avg Sales / Item	3.9 ★ Avg Customer Rating	8,523 Unique SKUs
10 Total Outlets	65% Low-Fat Items	T1–T3 Location Tiers	2011–2022 Establishment Range

Project Objective

Analyze Blinkit's grocery sales data to surface actionable insights on outlet performance, product category mix, customer satisfaction, and inventory visibility — enabling data-driven decisions around outlet expansion, product assortment, and operational efficiency.

Tools & Technologies

- MySQL 8.x — DDL schema design, DML data loading and cleaning
- SQL — Aggregate functions, GROUP BY, CTEs, Window Functions, Stored Views
- MySQL Workbench — Query execution and result visualization
- GitHub — Version control and project portfolio hosting

Dataset Overview

Source: Blinkit Grocery Sales Dataset (Kaggle)

Column	Type	Description
Item_Identifier	VARCHAR	Unique product code

Item_Weight	DECIMAL	Weight of the item
Item_Fat_Content	VARCHAR	Low Fat / Regular
Item_Visibility	DECIMAL	% shelf space (0–1)
Item_Type	VARCHAR	Product category (16 types)
Item_MRP	DECIMAL	Maximum Retail Price
Outlet_Identifier	VARCHAR	Unique outlet code
Outlet_Establishment_Year	INT	Year outlet opened
Outlet_Size	VARCHAR	Small / Medium / Large
Outlet_Location_Type	VARCHAR	Tier 1 / Tier 2 / Tier 3
Outlet_Type	VARCHAR	Supermarket Type 1/2/3, Grocery Store
Item_Outlet_Sales	DECIMAL	Target variable — revenue per item per outlet

Data Cleaning: Normalized Item_Fat_Content values — 'LF' and 'low fat' → 'Low Fat'; 'reg' → 'Regular' — using SQL UPDATE statements before analysis.

Business Questions & SQL Approach

Q1 — Total Sales, Avg Sales & Item Count

Objective: Establish baseline KPIs across the entire dataset.

Technique: SUM(), AVG(), COUNT() aggregate functions over the full table.

Q2 — Fat Content Impact on Sales

Objective: Determine whether health-conscious (low-fat) products drive more revenue.

Technique: GROUP BY Item_Fat_Content with SUM and AVG aggregations.

Q3 — Top Item Categories by Revenue

Objective: Identify which of the 16 product categories generate the most revenue.

Technique: GROUP BY Item_Type, ORDER BY Total_Sales DESC, LIMIT 10.

Q4 — Fat Content Breakdown Within Item Types

Objective: Cross-analyze health preference by product category.

Technique: Multi-level GROUP BY on Item_Type and Item_Fat_Content.

Q5 — Establishment Year vs. Sales

Objective: Understand how outlet maturity correlates with revenue.
Technique: GROUP BY Outlet_Establishment_Year, ORDER BY year for time-series view.

Q6 — Outlet Size Performance

Objective: Determine which outlet format (Small/Medium/Large) is most profitable.
Technique: GROUP BY Outlet_Size, SUM/AVG aggregation, RANK() window function.

Q7 — Full KPI Breakdown by Outlet Type

Objective: Master view of all KPIs segmented by Supermarket Type 1/2/3 and Grocery Store.
Technique: Multi-metric aggregation on Outlet_Type — the primary segmentation dimension.

Query Results — Key Outputs

Sales by Outlet Type (Q7)

Outlet Type	Items Sold	Total Sales	Avg Sales	Avg MRP
Supermarket Type1	2,388	\$787,549	\$149	\$143
Grocery Store	1,083	\$151,939	\$140	\$137
Supermarket Type3	935	\$130,714	\$140	\$141
Supermarket Type2	928	\$131,477	\$142	\$142

Sales by Outlet Size (Q6)

Outlet Size	Outlet Count	Total Sales	Avg Sales
Medium	6	\$507,907	\$142
Small	3	\$444,793	\$136
Large	1	\$248,987	\$149

Top 5 Item Categories by Revenue (Q3)

Item Type	Item Count	Total Sales	Avg Sales
Fruits and Vegetables	1,232	\$178,122	\$145
Snack Foods	1,200	\$175,433	\$146

Household	910	\$135,978	\$149
Frozen Foods	856	\$118,522	\$138
Dairy	682	\$114,603	\$168

Fat Content Analysis (Q2)

Fat Content	Item Count	Total Sales	Avg MRP
Low Fat	5,517	\$776,317	\$143
Regular	3,006	\$425,370	\$141

Business Insights & Recommendations

1. Supermarket Type 1 is the revenue engine

- Generates ~65% of total sales despite being one outlet type
- Recommendation: Prioritize stock depth, product variety, and shelf visibility at Type 1 outlets

2. Tier 3 Grocery Stores outperform Tier 3 Supermarkets

- Grocery stores in Tier 3 cities generate disproportionately high sales for their size
- Recommendation: Expand Grocery Store footprint in Tier 3 cities before investing in Supermarkets

3. Medium outlets deliver near-Large revenue with lower overhead

- Medium: \$507K total sales vs Large: \$248K — with 6 vs 1 outlet count
- Average sales per item: Medium (\$142) closely tracks Large (\$149)
- Recommendation: Medium-format expansion offers better capital efficiency than Large stores

4. Low-Fat products dominate — optimize procurement accordingly

- 65% of SKUs and 58% of revenue are Low-Fat products
- Recommendation: Increase Low-Fat SKU assortment, especially in health-conscious Tier 1 markets

5. Customer ratings are uniform — compete on ops, not rating campaigns

- All outlet types rate between 3.8–4.0 — statistically indistinguishable

- Recommendation: Improve delivery speed, stock availability, and pricing transparency instead

6. 2018 was peak vintage — monitor post-2020 outlet ramp-up

- Outlets established in 2018 show the highest average sales
- Post-2020 outlets are still in early stages — set 18-month ramp benchmarks

SQL Code Highlights

Schema creation (DDL)

```
CREATE TABLE blinkit_sales (
  Item_Identifier VARCHAR(20),
  Item_Fat_Content VARCHAR(20),
  Item_Type VARCHAR(50),
  Item_Outlet_Sales DECIMAL(10,2),
  Outlet_Type VARCHAR(50), ...);
```

Window function — rank outlets within tier

```
SELECT Outlet_Identifier, Outlet_Location_Type, Total_Sales,
  RANK() OVER (
    PARTITION BY Outlet_Location_Type
    ORDER BY Total_Sales DESC
  ) AS Rank_Within_Tier
FROM vw_outlet_summary;
```

GitHub Repository Structure

File / Folder	Purpose
sql/create_database.sql	Schema setup, LOAD DATA INFILE, data cleaning
sql/kpi_queries.sql	All 7 business questions + 2 bonus analyses
sql/advanced_analysis.sql	CTEs, Window Functions, CREATE VIEW
docs/Blinkit_Sales_Analysis.docx	This submission document
README.md	Project overview, setup instructions, findings
screenshots/	Optional: query output screenshots

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